

João Renato Leripio

Statistical Modeling, Forecasting & Data Science

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Profile

Senior data science and modeling professional specializing in forecasting, time series analysis, and alternative and unstructured data. Combines hands-on statistical modeling and technical leadership with the development of reliable analytical pipelines and data products for economic and financial research.

Areas of Expertise

Data: Alternative, Unstructured, and High-Frequency Data

Modeling: Econometrics · Machine Learning · State-Space Models · Bayesian Inference

Delivery: Data Pipelines · APIs · Data Apps · Automated Reporting

Professional Experience

Kapitalo Investimentos

Data Science & Economic Modeling Lead

São Paulo, Brazil · 2022–Present

- Lead the Data Science and Modeling team within Macroeconomic Research, providing technical direction and overseeing the development of quantitative models and analytical products.
- Develop forecasting and nowcasting models for inflation and economic activity using alternative, high-frequency, and unstructured data.
- Design and implement models to estimate latent economic variables, including potential output and the NAIRU, at scale across dozens of countries.
- Build end-to-end analytical workflows that operationalize models and research, from data collection and estimation to automated reporting and internal applications.

Itaú Asset Management

Quantitative Research Analyst

São Paulo, Brazil · 2019–2022

- Developed economic activity and inflation indicators using alternative and unstructured data, expanding the information available for macroeconomic analysis.
- Designed and implemented forecasting models for COVID-19 cases and vaccination coverage, developing modeling approaches for rapidly evolving processes with limited historical data.
- Developed forecasting models for revenues of publicly listed companies, combining economic and company-level data to support investment research.

- Built machine learning models using credit-market data to predict corporate bond prices.

IPEDF

Economic Research Analyst

Brasília, Brazil · 2017–2019

- Conducted applied economic research and monitored regional economic activity using official and administrative data.
- Developed statistical indicators and analytical tools to measure economic conditions and support public-policy discussions.
- Produced reports and presented quantitative findings to technical and nontechnical audiences.

Education

M.S. in Economics, Universidade Federal Fluminense · 2017

B.S. in Economics, Universidade Federal Fluminense · 2013

Publication

Author of *R for Economic Research*, a practical guide to data workflows, statistical modeling, time series, and forecasting for applied research. The book has received more than 50,000 views from readers in over 50 countries.

Selected Presentations

- **Nowcasting When Predictors Keep Changing: Leveraging Cross-Sectional Information in High-Dimensional, High-Turnover Data** — International Symposium on Forecasting · Montreal · 2026
- **Nowcasting Inflation in Brazil Using Web Data** — International Symposium on Forecasting · Charlottesville · 2023
- **Real-Time Forecasting of COVID-19 Cases Across US Regions** — International Symposium on Forecasting · Remote · 2020